EXP-4 : Create a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

1. Aim

To create a visual representation of mobile network protocol stacks using Figma, illustrating the different protocol layers and their roles in communication between a mobile device, radio access network, core network, and external networks.

2. Software Used

- Figma: For designing the protocol stack diagram and visualizing network components.
- Web Browser: To access and use Figma.

3. Algorithm

- Start.
- Open Figma and create a new design file.
- Create four sections: User Equipment, Radio Access Network, Core Network, and External Networks.
- Represent each section using a separate frame or panel.
- Add the Application, Transport, Network, Data Link, and Physical layers.
- Include relevant protocols such as HTTP/HTTPS, TCP/UDP, IP, RLC/MAC, RRC, NAS, SCTP, and GTP.
- Connect the network sections using arrows to represent communication interfaces.
- Add labels, icons, colors, and descriptions for each layer.
- Arrange the components neatly and review the completed diagram.
- End.

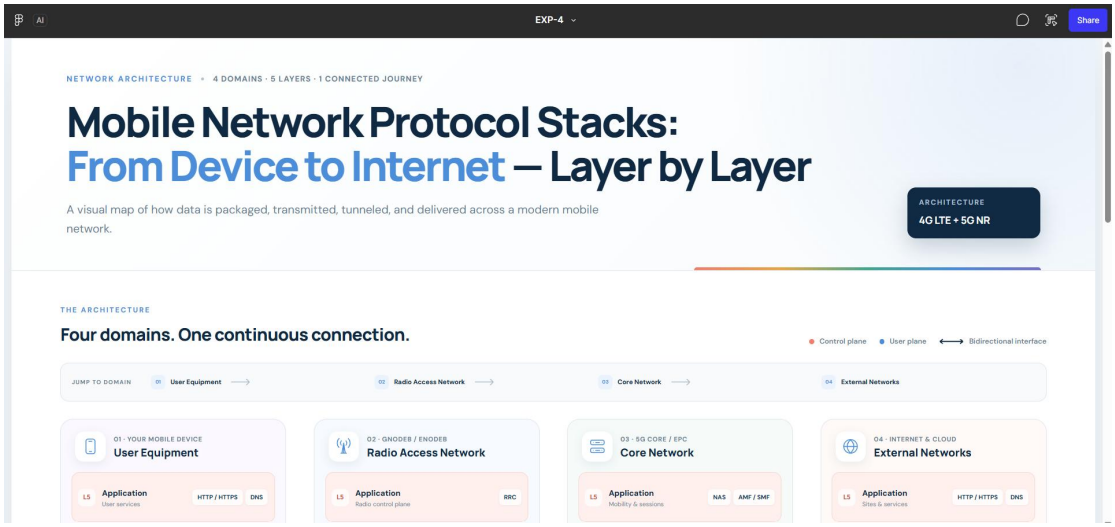
4. Design

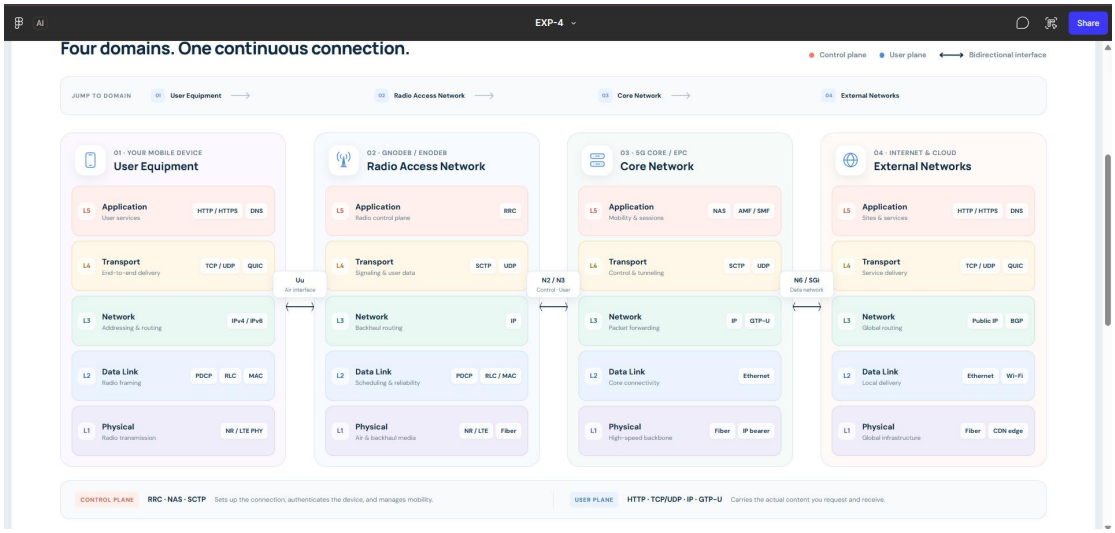
The design uses a layered structure to explain how information travels through a mobile network.

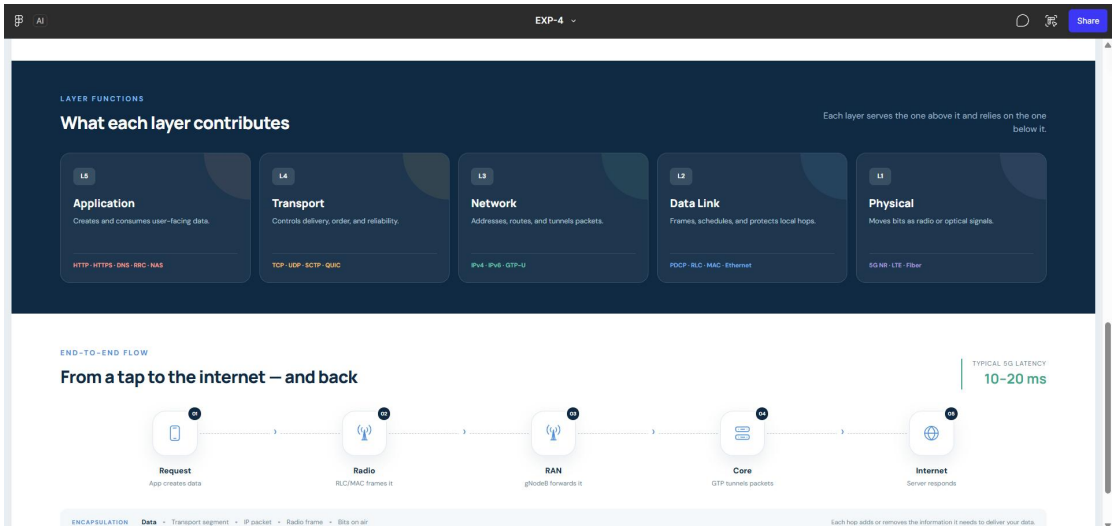
- User Equipment (UE): Represents a mobile phone or device.
- Radio Access Network (RAN): Provides wireless connectivity through BTS and BSC.
- Core Network (CN): Manages switching, subscriber information, mobility, and routing.

- External Networks: Represents the internet and other connected services.

Each network section contains protocol layers, distinguished by different colors. Arrows between the sections represent interfaces and communication paths.

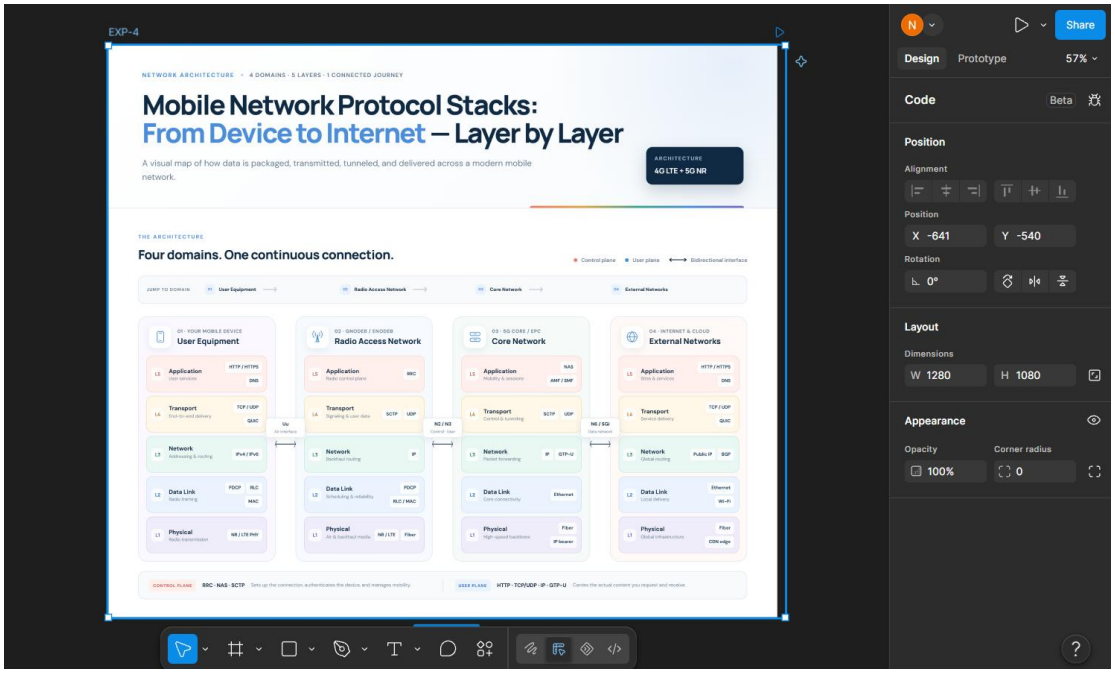






5. Prototype

The prototype is a single visual diagram containing the four network sections, their protocol layers, key protocols, and connecting arrows.



Result:

Thus, a visual representation of mobile network protocol stacks was designed using Figma. The diagram illustrates the protocol layers, key network components, and communication interfaces involved in transferring data from a mobile device to external networks.